Fundamentals of Solid State Physics

Origin of Optical Properties

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Optical Properties of Materials

Metal

SiO_2

Silicon

- **Crystal Structures**
  - polycrystalline, amorphous, single crystalline
- **Electronics**
  - conductor, insulator, semiconductor
- **Optics (in the visible range)**
  - reflective, transparent, absorbing

Outline

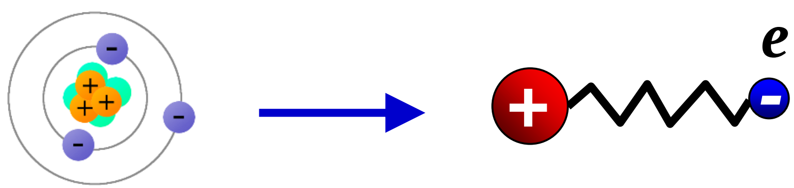
$$\boxed{\tilde{n} = n + i\kappa}$$

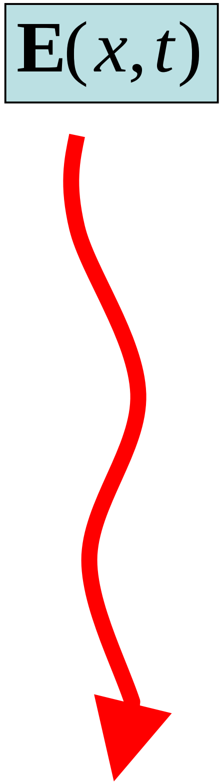
- **Refractive index**
  - **dipole polarization - oscillator model**

- **Absorption**
  - **damped oscillator**
  - **free carriers, band transitions, optical phonons, defects, ...**

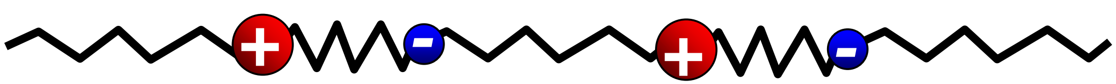
Origin of ε_r and $\tilde{n}$

- **Interaction between EM wave and charges (electrons, ions, etc.) in the solids**
  - **Oscillation between electrons and nuclei**

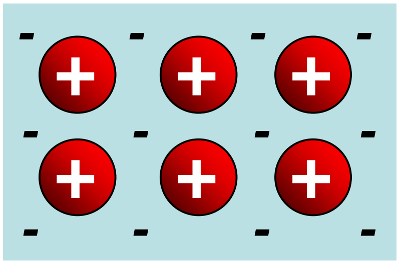




  - **Oscillation between ions (phonon)**

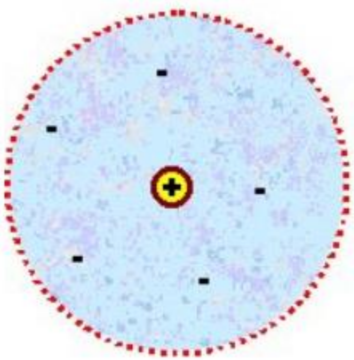


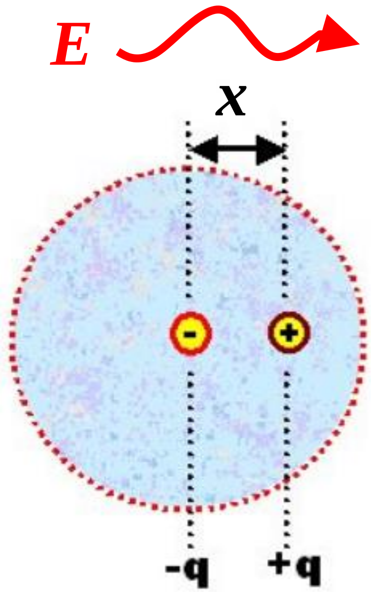
  - **Oscillation of free electron gas**



The Dipole Polarization

atom





Polarization (极化)

$$\boxed{\mathbf{P} = nq\mathbf{x}}$$

n - density of dipoles
q - unit charge
x - displacement

$$\begin{aligned}\mathbf{D} &= \varepsilon_0\mathbf{E} + \mathbf{P} \\ &= \varepsilon_0\mathbf{E} + \sum \mathbf{P}_{\text{dipole}} \\ &= \varepsilon_0\mathbf{E} + \sum nq\mathbf{x}\end{aligned}$$

$$\boxed{\mathbf{D} = \varepsilon_0\varepsilon_r\mathbf{E}}$$

$$\longrightarrow \quad \boxed{\varepsilon_r = 1 + \sum \frac{nq\mathbf{x}}{\varepsilon_0\mathbf{E}}}$$

dielectric constant = vacuum + all the dipoles

The Dipole Oscillator Model

$\mathbf{E}(x,t)$

oscillation frequency $\omega_0 = \sqrt{\dfrac{K}{m}}$

$$m\frac{dv}{dt} + m\frac{v}{\tau} + Kx = eE(t)$$

Lorentz Model

$$\rightarrow \quad m\frac{d^2x}{dt^2} + \frac{m}{\tau}\frac{dx}{dt} + m\omega_0^{\ 2}x = eE(t)$$

acceleration **damping relaxation** **Spring force** **Electric force**

The Dipole Oscillator Model

Lorentz Model

$$m\frac{d^2x}{dt^2}+\frac{m}{\tau}\frac{dx}{dt}+m\omega_0^2x=eE(t)$$

$$x=x_0e^{-i\omega t}$$

$$\rightarrow \quad x=\frac{e}{m(\omega_0^2-\omega^2-i\omega/\tau)}E$$

$$\rightarrow \quad \varepsilon_r=1+\frac{nq\mathbf{x}}{\varepsilon_0\mathbf{E}}=1+\frac{ne^2}{\varepsilon_0 m}\cdot\frac{1}{\omega_0^2-\omega^2-i\omega/\tau}$$

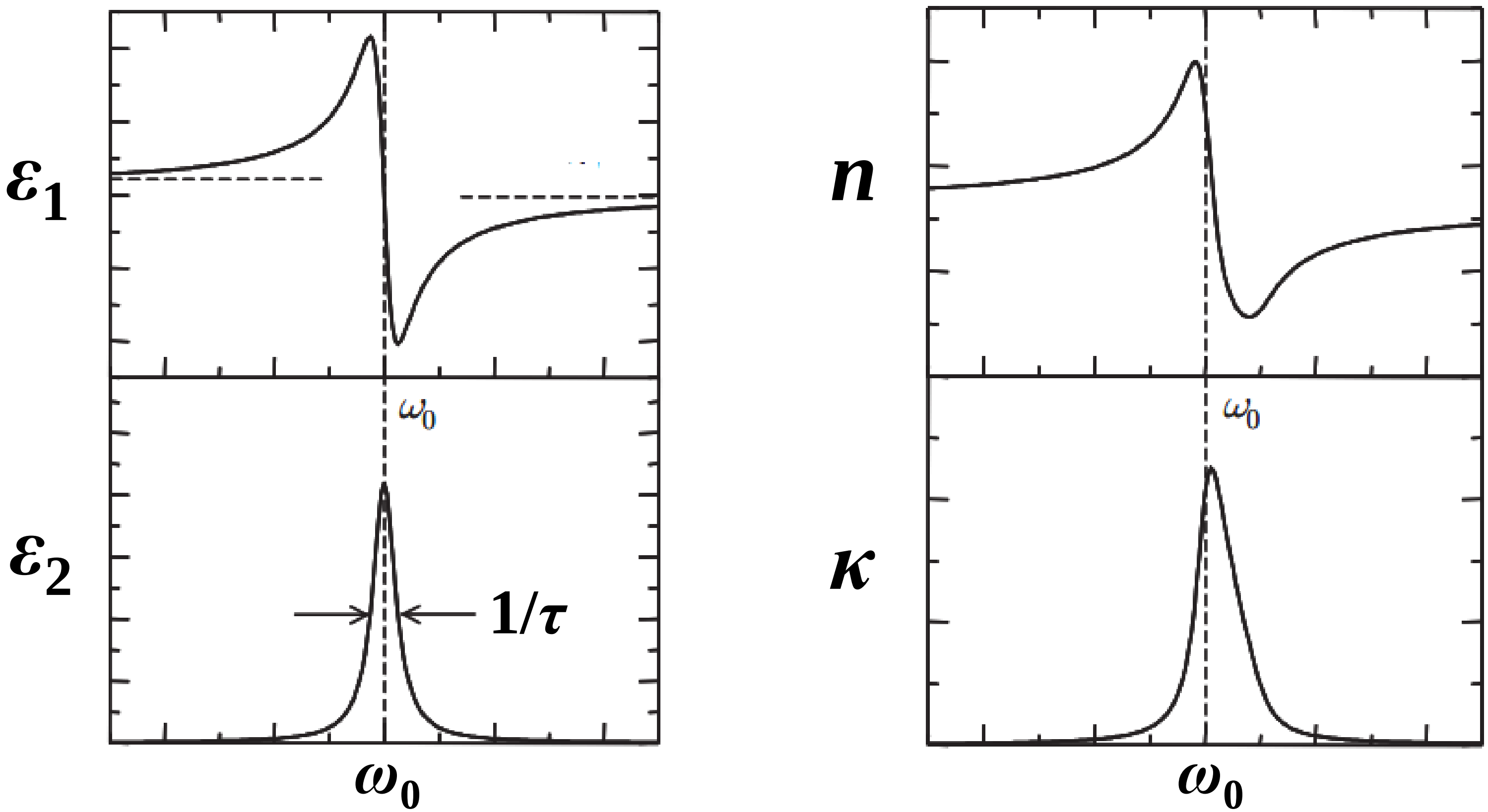
contribution of one resonance

The Dipole Oscillator Model

$$\tilde{\varepsilon}_r = \varepsilon_1 + i\varepsilon_2$$

$$\tilde{n} = \sqrt{\tilde{\varepsilon}_r} = n + i\kappa$$

resonance at ω_0

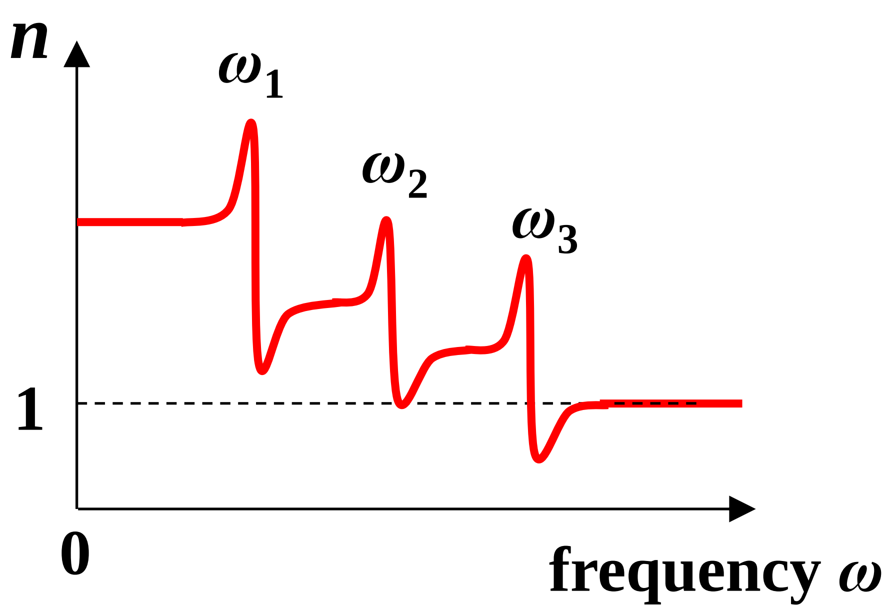


The Dipole Oscillator Model

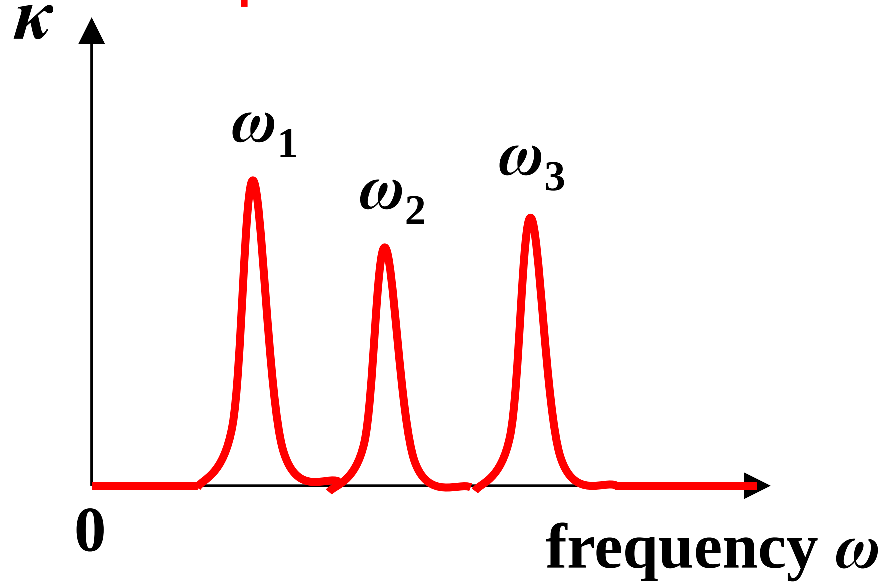
Multiple resonances

$$\varepsilon_r = 1 + \sum \frac{nq\mathbf{x}}{\varepsilon_0 \mathbf{E}}$$

$$= 1 + \frac{ne^2}{\varepsilon_0 m} \cdot \sum_j \frac{1}{\omega_j^2 - \omega^2 - i\omega/\tau_j}$$



absorption

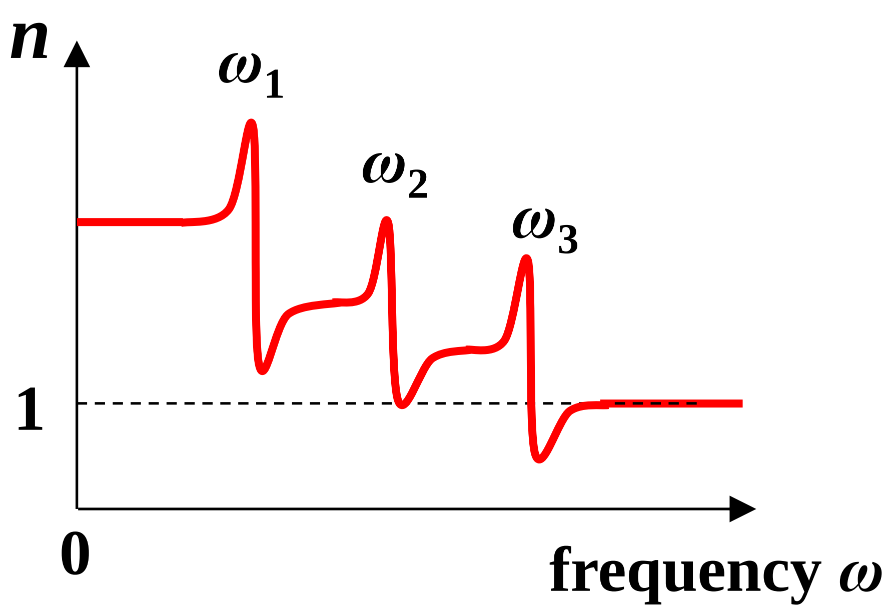


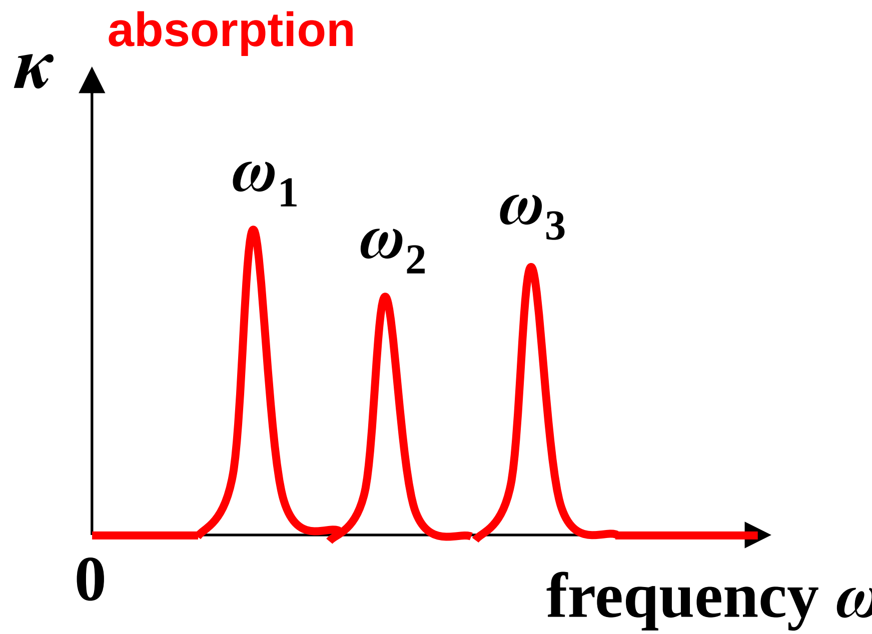
The Dipole Oscillator Model

When $\omega \approx 0$, n and ε_r is constant (dielectric) in DC field

When ω is at resonance, strong absorption

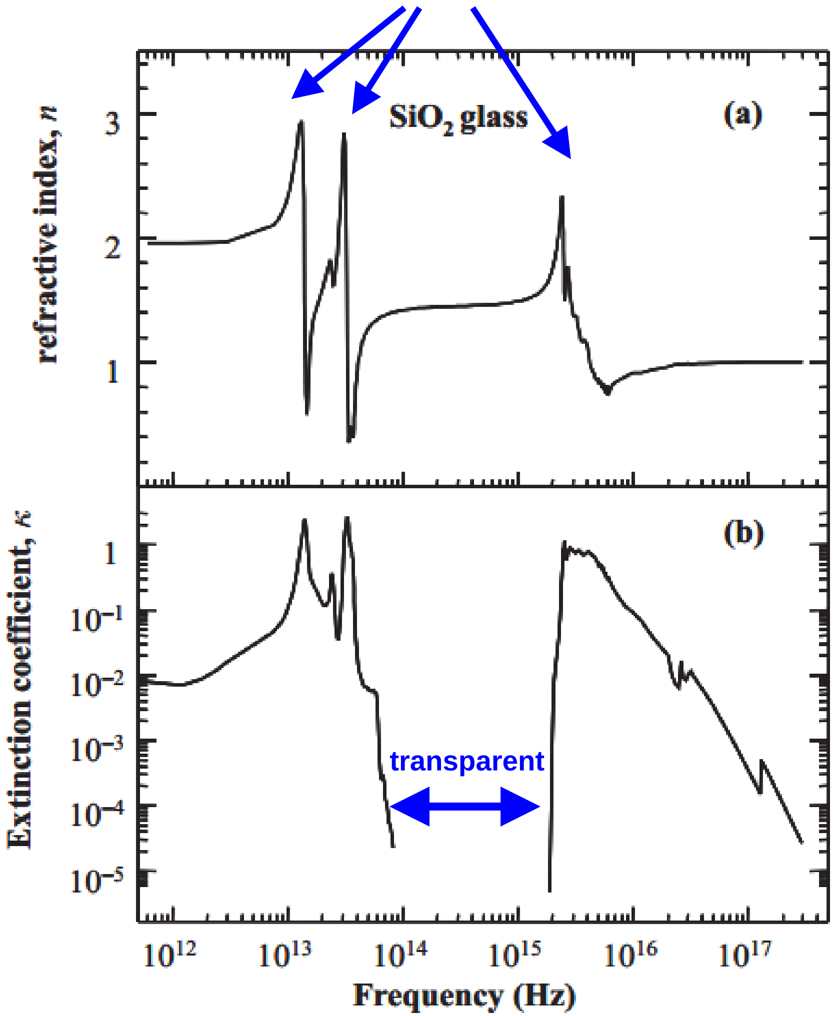
When $\omega = +\infty$, $n = 1$, $\kappa = 0$. Transparent like vacuum (High frequency x-rays and γ-rays can penetrate most materials)





Example: SiO_2 glass

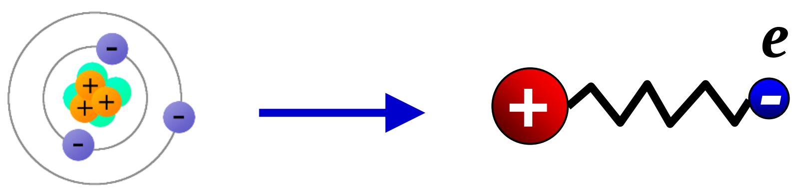
dipole resonances

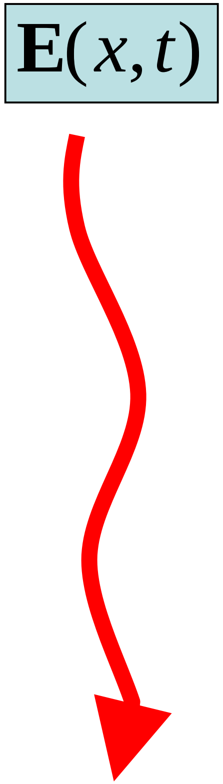


glass is transparent from 300 nm to 3000 nm

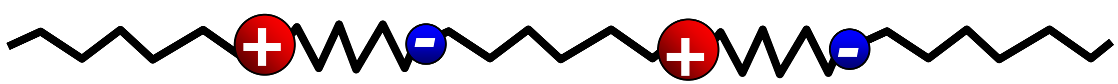
Origin of ε_r and $\tilde{n}$

- **Interaction between EM wave and charges (electrons, ions, etc.) in the solids**
  - **Oscillation between electrons and nuclei**

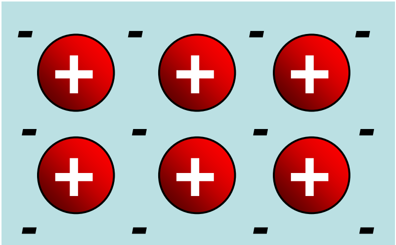




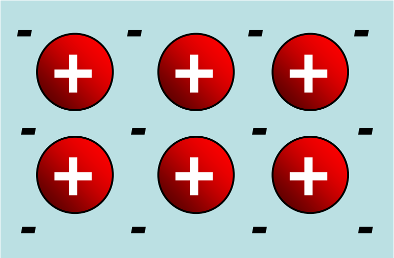
  - **Oscillation between ions (phonon)**



  - **Oscillation of free electron gas**



Free Electrons in Metals



positive ions
+
electron cloud

The Drude Model: Free electron 'gas'

- **Independent**
  - electrons do not interact with each other
- **Free**
  - electrons do not interact with ions, except collision
- **Collision**
  - electrons are scattered by the ions instantaneously
- **Relaxation time τ 弛豫时间**
  - average time between two collisions
  - electron mean free path $l = v^*\tau$
- **Maxwell–Boltzmann distribution**
  - average kinetic energy

$$\boxed{\frac{1}{2}mv^2 = \frac{3}{2}k_B T}$$

P. Drude
1863–1906

Free Electrons in Metals

Drude-Lorentz Model

$$\mathbf{F} = m\frac{d\mathbf{v}}{dt} + m\frac{\mathbf{v}}{\tau} = e\mathbf{E}(t)$$

τ - relaxation time (s) 弛豫时间

when E is constant, v is constant

$$\rightarrow \quad \frac{d\mathbf{v}}{dt} = 0 \quad \rightarrow \quad m\frac{\mathbf{v}}{\tau} = e\mathbf{E}$$

$$\mu = \frac{\mathbf{v}}{\mathbf{E}} = e\frac{\tau}{m} \qquad \sigma = ne\mu = ne^2\frac{\tau}{m} \qquad \mathbf{j} = ne\mathbf{v} = \sigma\mathbf{E}$$

mobility | conductivity | Ohm's law

Free Electrons in Metals

Drude-Lorentz Model

$$\mathbf{F} = m\frac{d\mathbf{v}}{dt} + m\frac{\mathbf{v}}{\tau} = e\mathbf{E}(t)$$

when interacting with AC field (Optical wave)

$$m\frac{d^2x}{dt^2} + \frac{m}{\tau}\frac{dx}{dt} = eE(t)$$

$$x = x_0 e^{-i\omega t} \longrightarrow x = -\frac{e}{m(\omega^2 + i\omega/\tau)}E$$

$$\longrightarrow \varepsilon_r = 1 + \frac{nq\mathbf{x}}{\varepsilon_0\mathbf{E}} = 1 - \frac{ne^2}{\varepsilon_0 m}\cdot\frac{1}{\omega^2 + i\omega/\tau}$$

Free Electrons in Metals

$$\varepsilon_r = 1 + \frac{nq\mathbf{x}}{\varepsilon_0 \mathbf{E}} = 1 - \frac{ne^2}{\varepsilon_0 m} \cdot \frac{1}{\omega^2 + i\omega/\tau}$$

For a weakly damp system, 1/τ ≈ 0

$$\rightarrow \quad \varepsilon_r = 1 - \frac{ne^2}{\varepsilon_0 m} \cdot \frac{1}{\omega^2} = 1 - \frac{\omega_p{}^2}{\omega^2}$$

$$\omega_p = \sqrt{\frac{ne^2}{\varepsilon_0 m}}$$

plasma frequency (rad/s)

Plasma frequency (ω_p) represents the oscillation of the whole electron gas in the solid.

Free Electrons in Metals

$$\tilde{n} = \sqrt{\varepsilon_r} = \sqrt{1 - \frac{\omega_p^{\,2}}{\omega^2}}$$

$$R = \left| \frac{\tilde{n} - 1}{\tilde{n} + 1} \right|^2$$

When $\omega < \omega_p$, $\tilde{n}$ is purely imaginary. R = 100%.

Metals are like a mirror at low frequency.

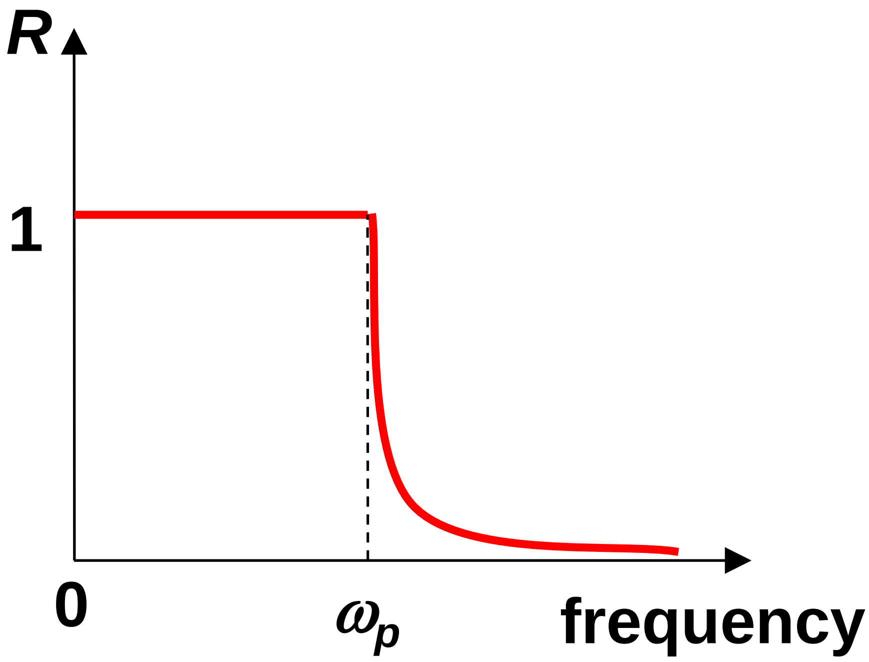
When $\omega > \omega_p$, $\tilde{n}$ is real. R decreases when ω increases
When $\omega = +\infty$, $\tilde{n} = 1$, $R = 0$. Transparent like vacuum

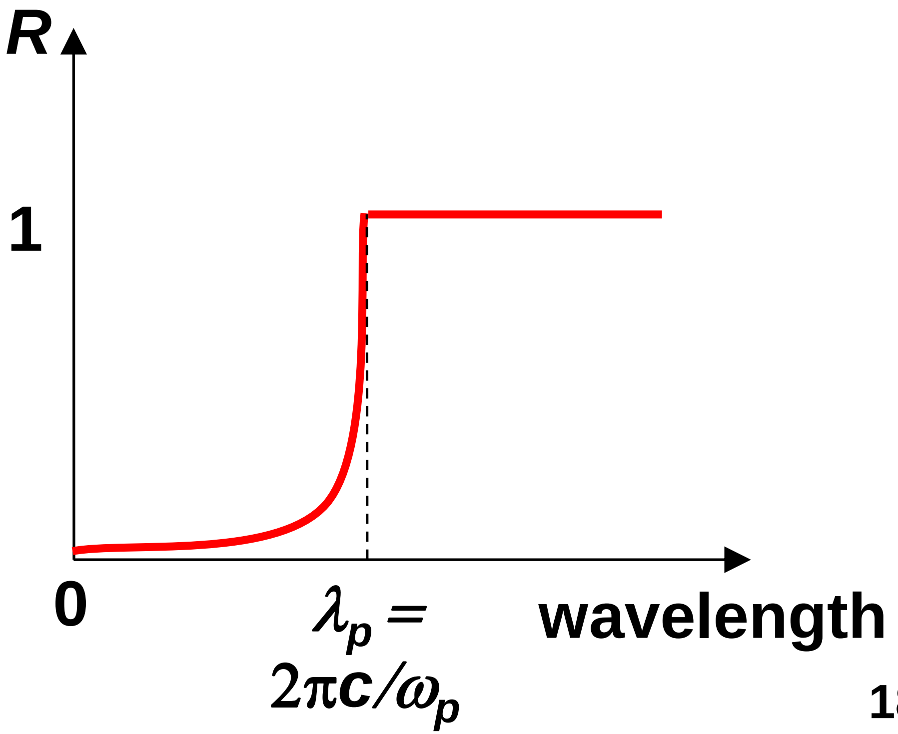
(High frequency x-rays and γ-rays can penetrate most materials)

Reflectivities of Metals

$$\tilde{n} = \sqrt{\varepsilon_r} = \sqrt{1 - \frac{\omega_p^2}{\omega^2}}$$

$$R = \left| \frac{\tilde{n} - 1}{\tilde{n} + 1} \right|^2$$

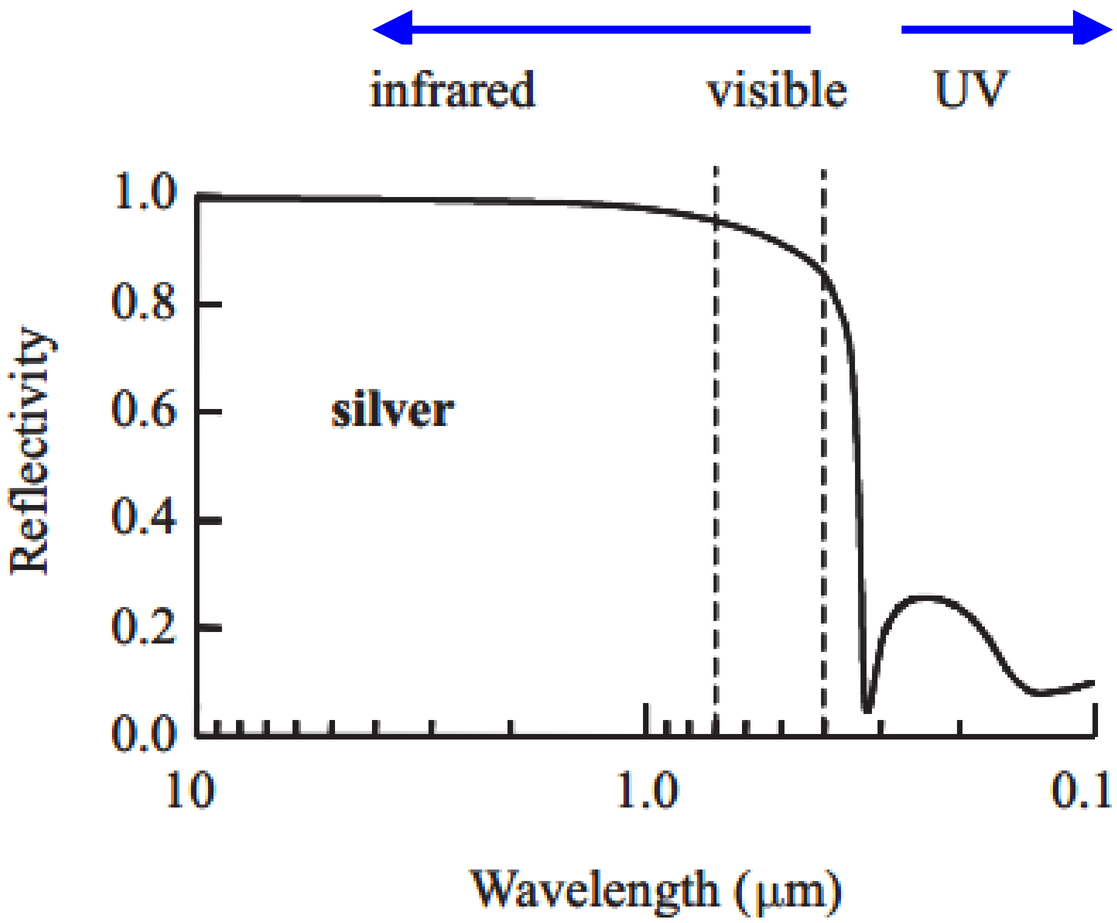




Example: Silver

reflection at long wavelength (infrared and visible)

Transmission at short wavelength (ultraviolet)



Silver mirror

Example: Metals

transmission at high frequency

reflection at low frequency

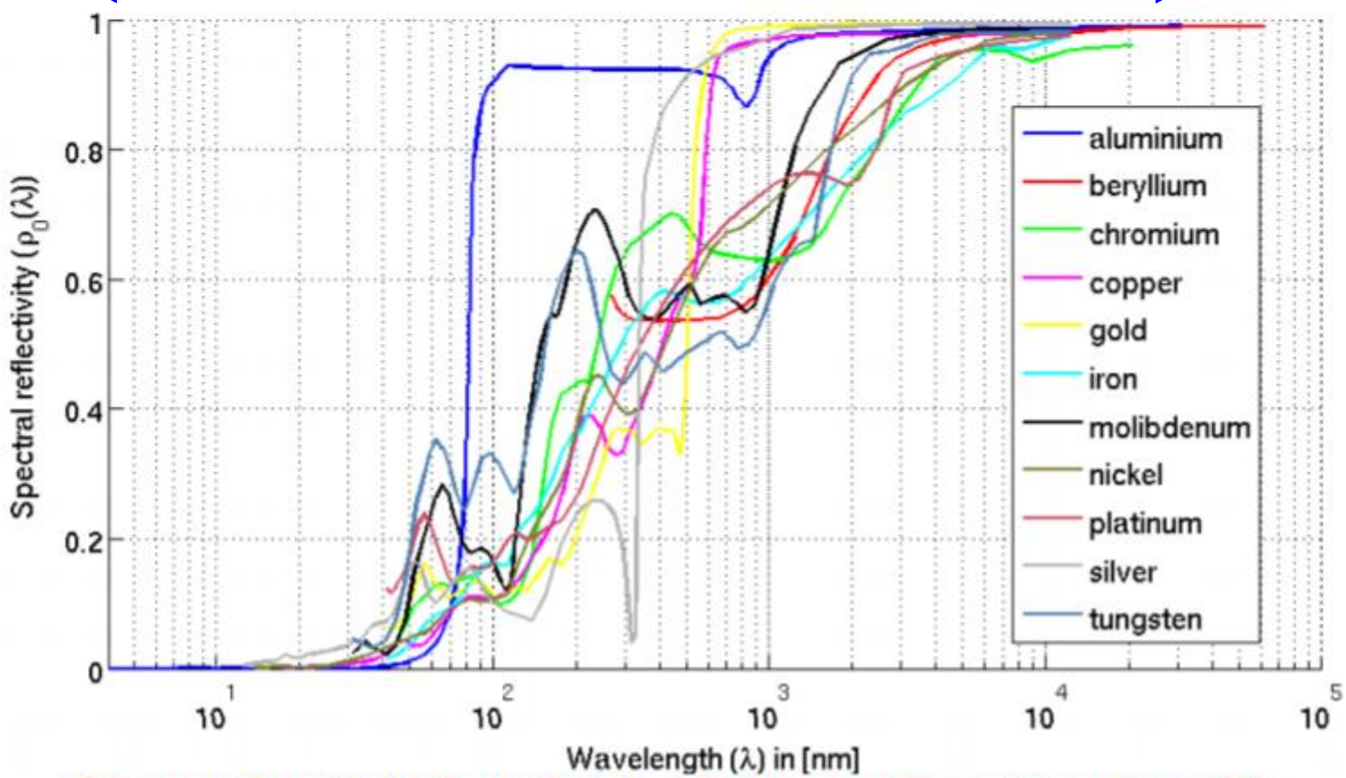


Figure 1: Spectral reflectivity of perfectly smooth metal surfaces [3]

https://doi.org/10.3929/ethz-a-006206911

Example: Metals

transmission at high frequency ←

→ **reflection at low frequency**

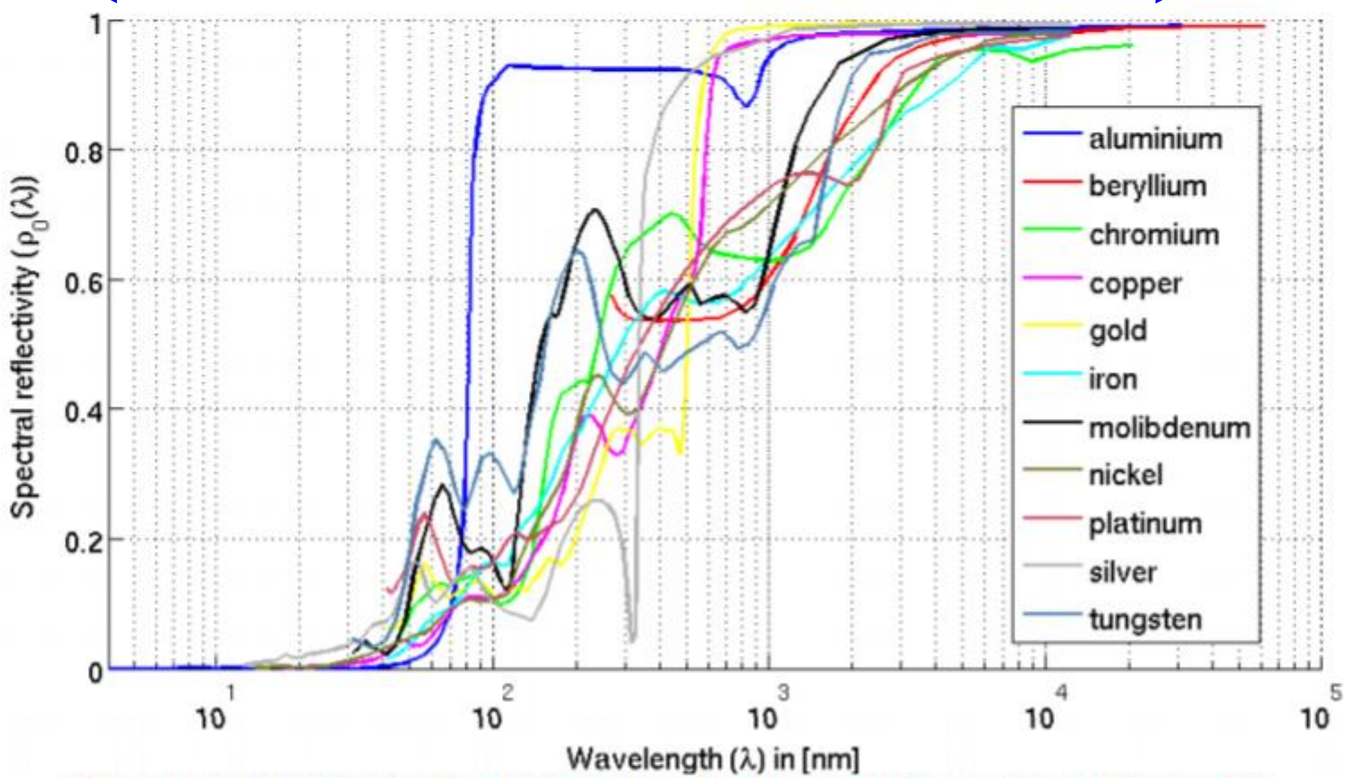
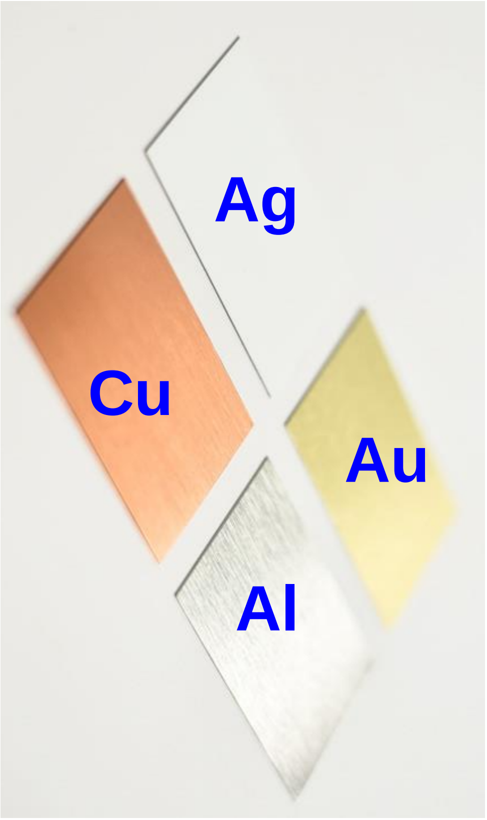


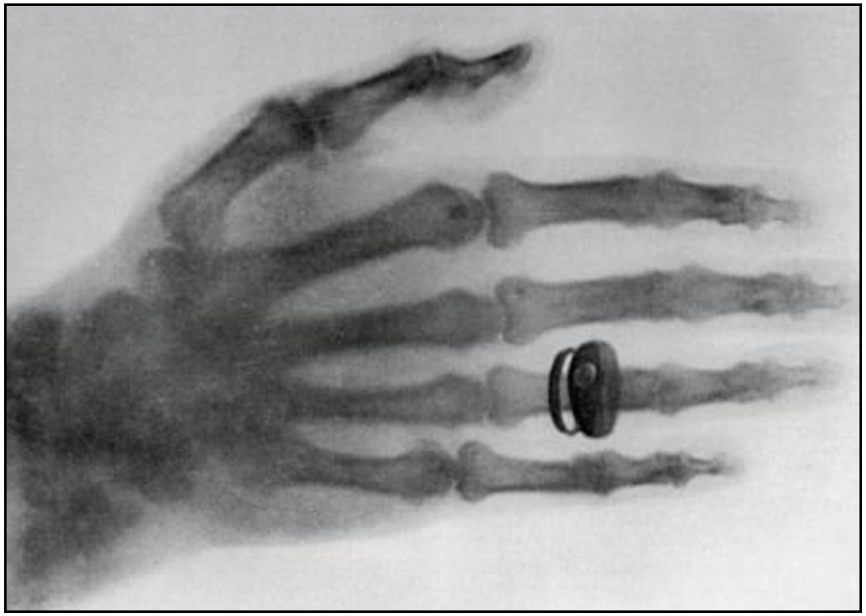
Figure 1: Spectral reflectivity of perfectly smooth metal surfaces [3]



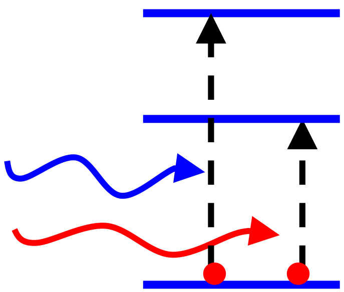
https://doi.org/10.3929/ethz-a-006206911

Example: X-ray Transmission

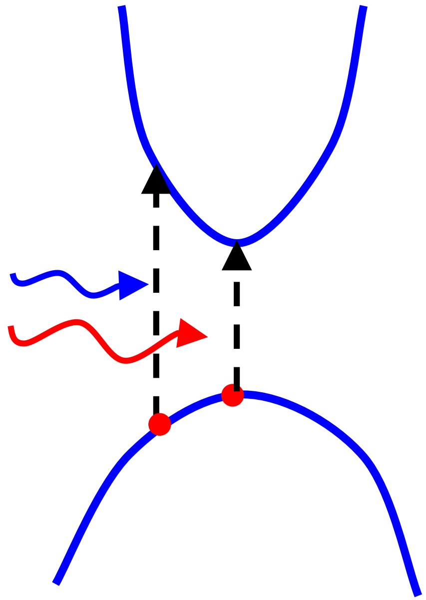
- **X-ray has higher transmission for light atoms (water, skin, fat, etc.)**
- **X-ray has higher absorption and reflection for heavy atoms (bones, metals, etc.)**



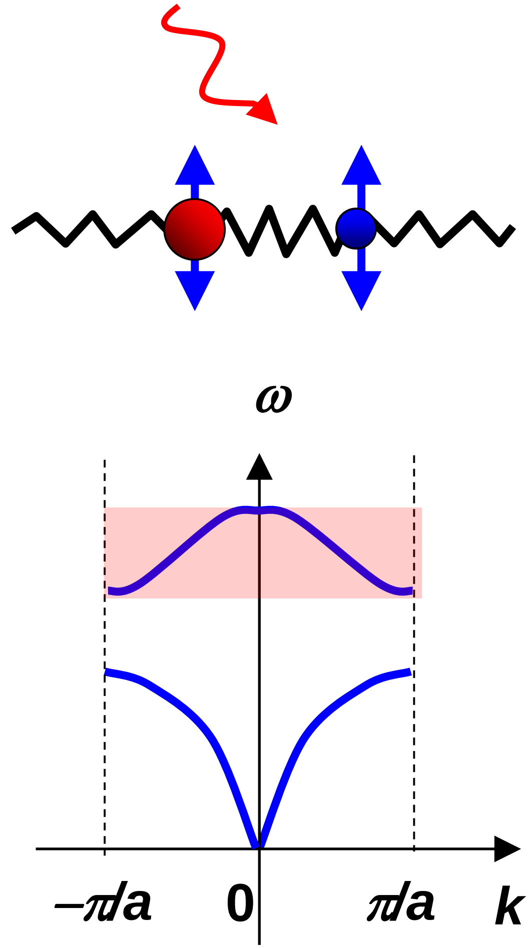
Origin of Optical Absorption κ



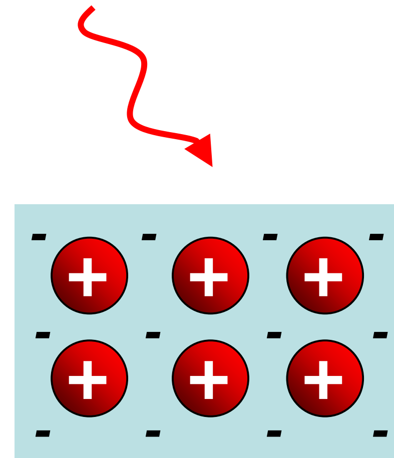
energy levels
(atoms,
defects, ...)



above band gap
(electronic
band theory)



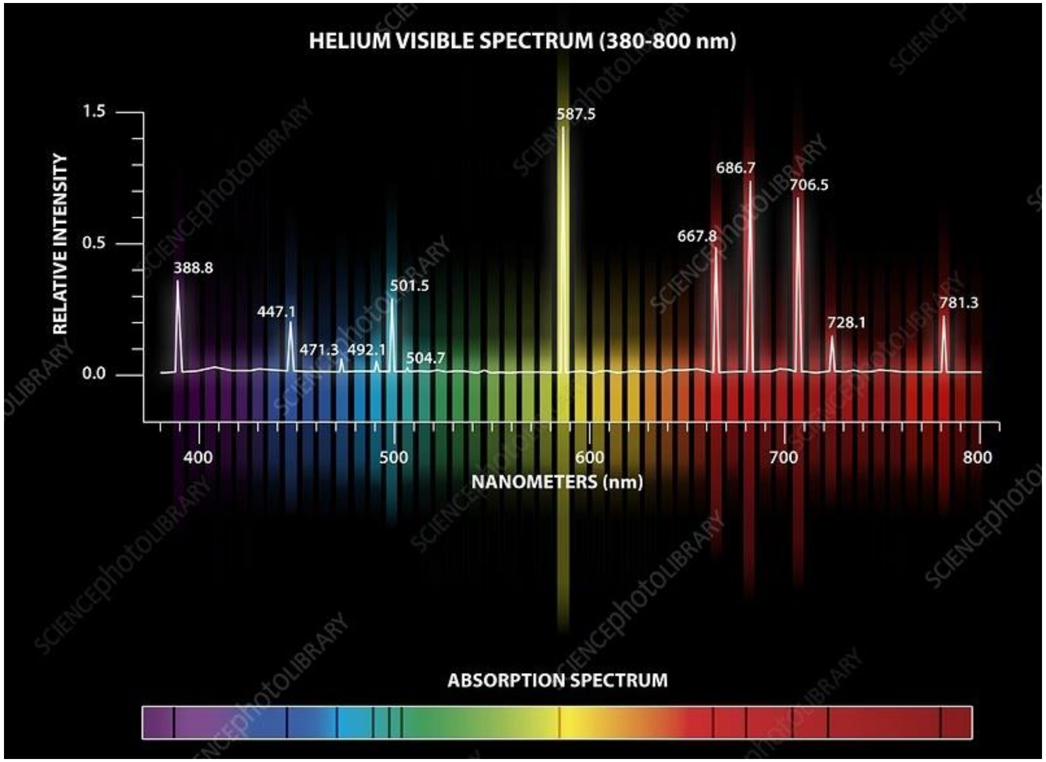
optical phonons
(Lorentz Model)

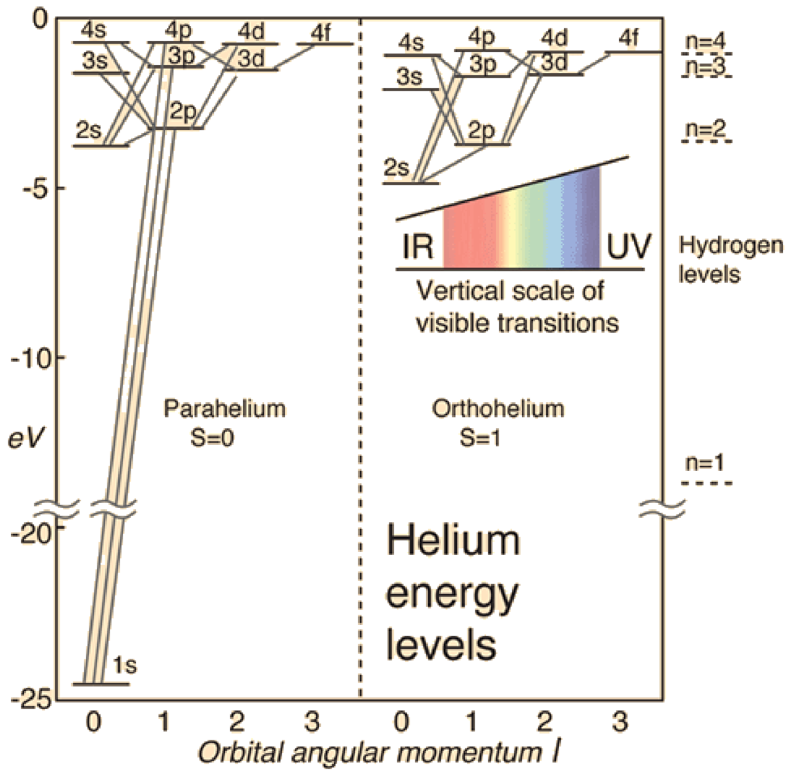


free carriers
(Drude Model)

Example: Helium

absorption lines are from energy transitions between discrete orbitals





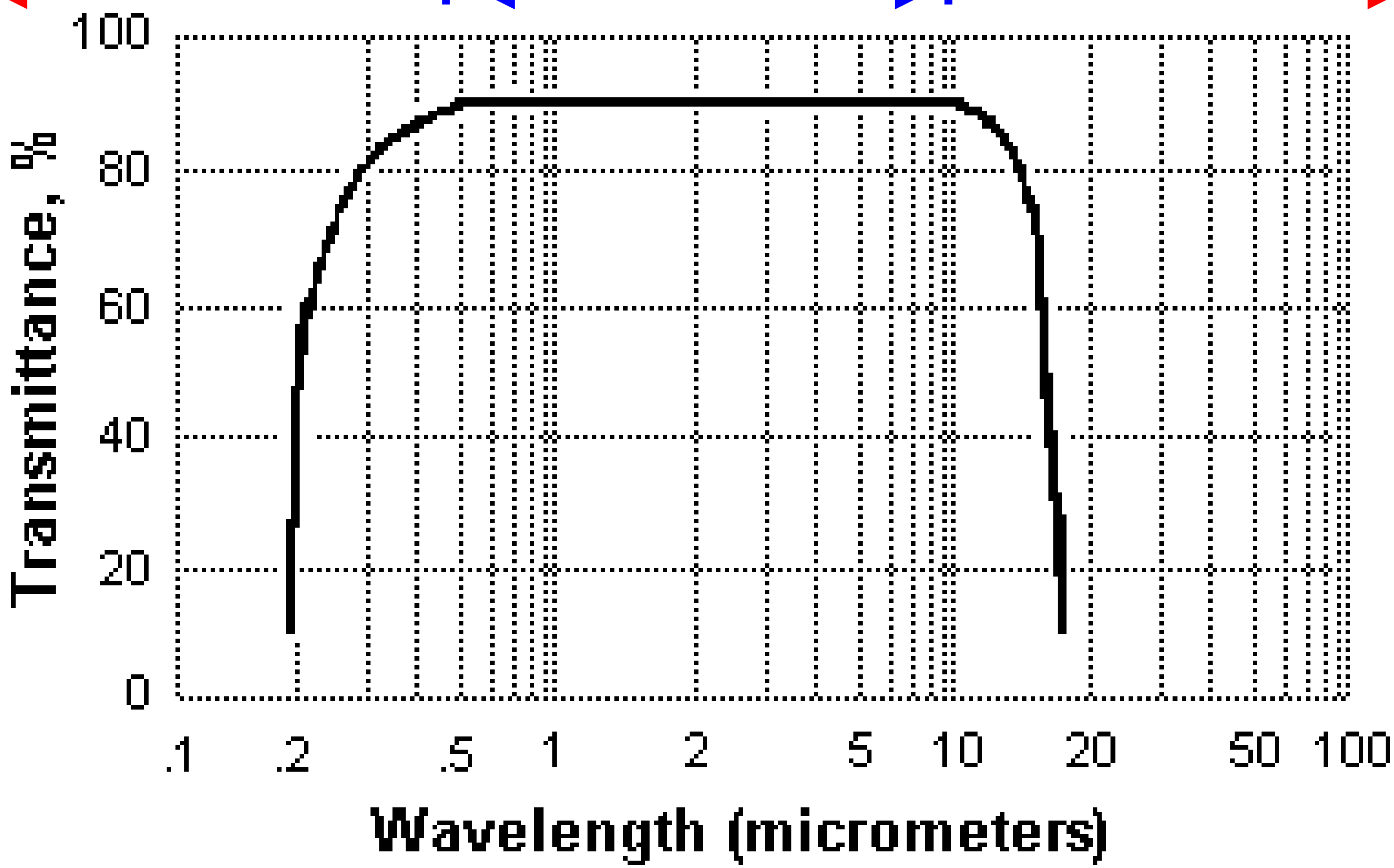
https://www.sciencephoto.com/media/673903/view/helium-emission-and-absorption-spectra
http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/helium.html

Example: NaCl

absorption above band gap
(E_g ~ 8.5 eV)

transparent

absorption by crystal vibration
($2\pi\omega_{max}/c$ ~ 10 um)

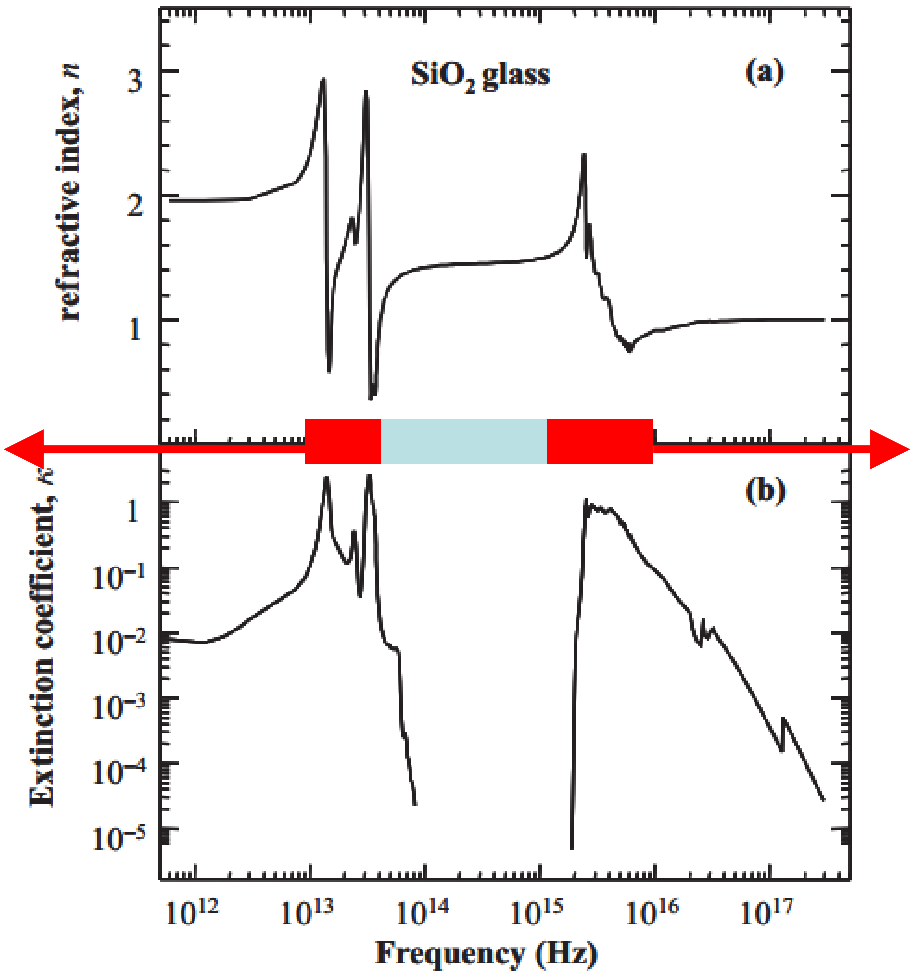


http://www.crystaltechno.com/NaCl_en.htm

Example: SiO_2 glass

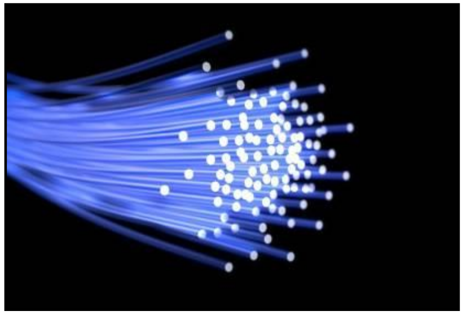
transparent in the visible and infrared (from 300 nm to 3000 nm)
($n \approx 1.4$)

infrared absorption of optical phonons (crystal vibration)

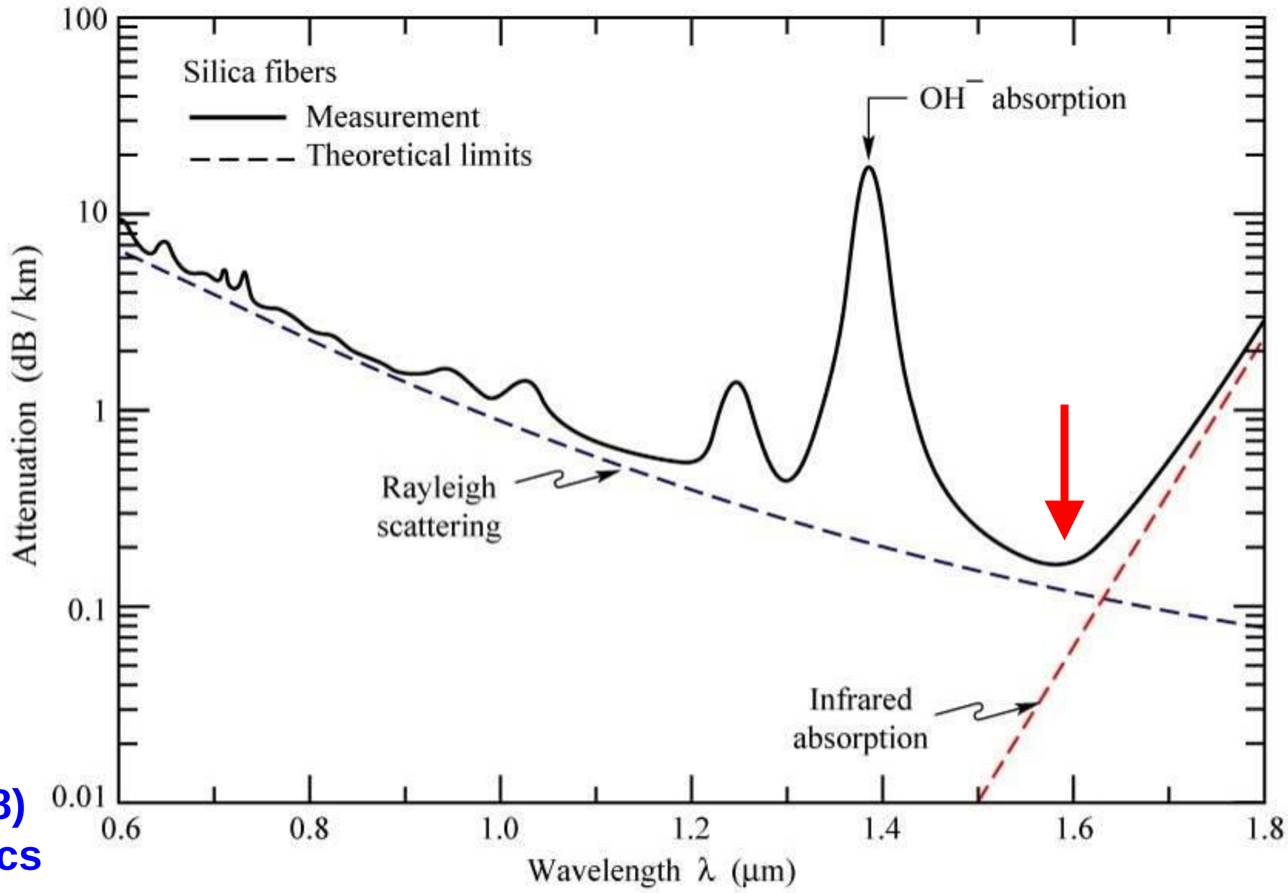


ultraviolet absorption above band gap (E_g ~ 9 eV)

Pure SiO_2 - Optical Fibers



K. Kao (高锟) (1933–2018)
2009 Nobel Prize in Physics



minimum loss at 1550 nm, 0.2 dB/km
~ 2% loss every kilometer

K. C. Kao, G. A. Hockham, *Proc. IEE* **113,** 1151 (1966)

Impurities in SiO_2

- **Why is the desert yellow?**
  - **because of Fe_2O_3 (Fe^{3+} ions)**

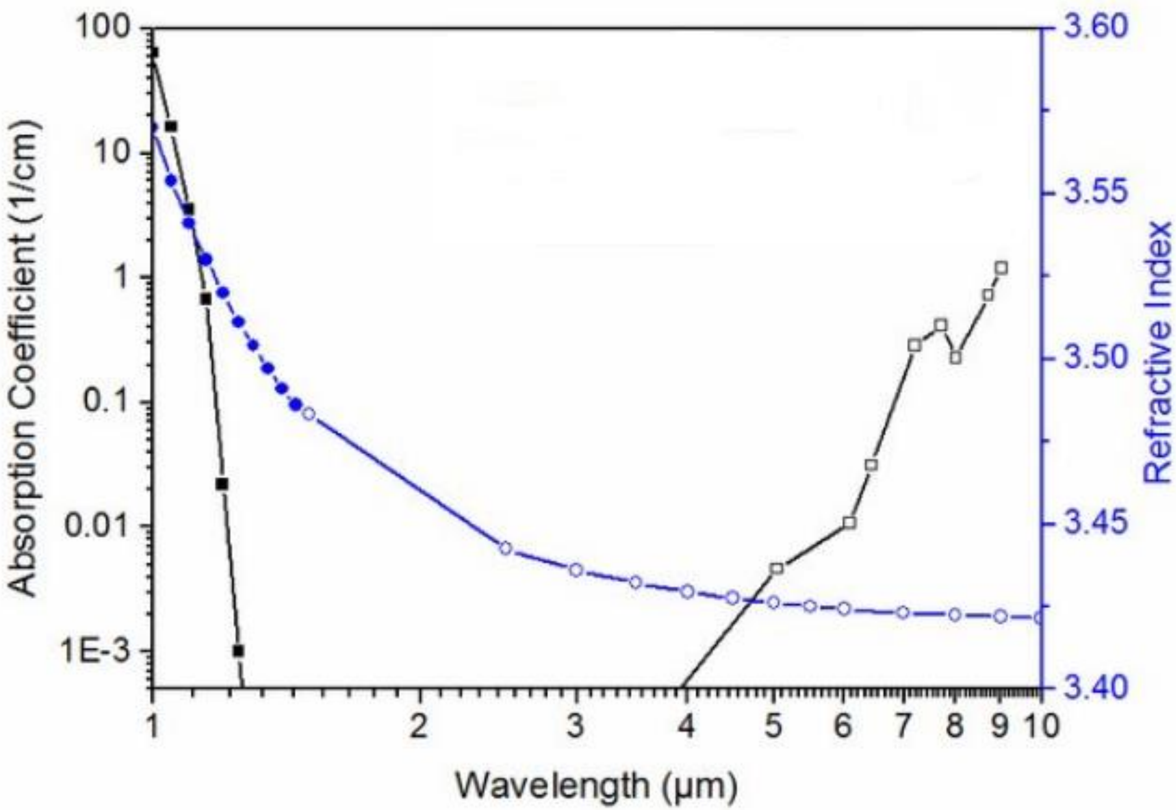
- **Why are beer bottles green?**
  - **because of FeO (Fe^{2+} ions)**

Example: Silicon (intrinsic)

absorption above band gap (E_g = 1.1 eV) (λ < 1.2 μm)

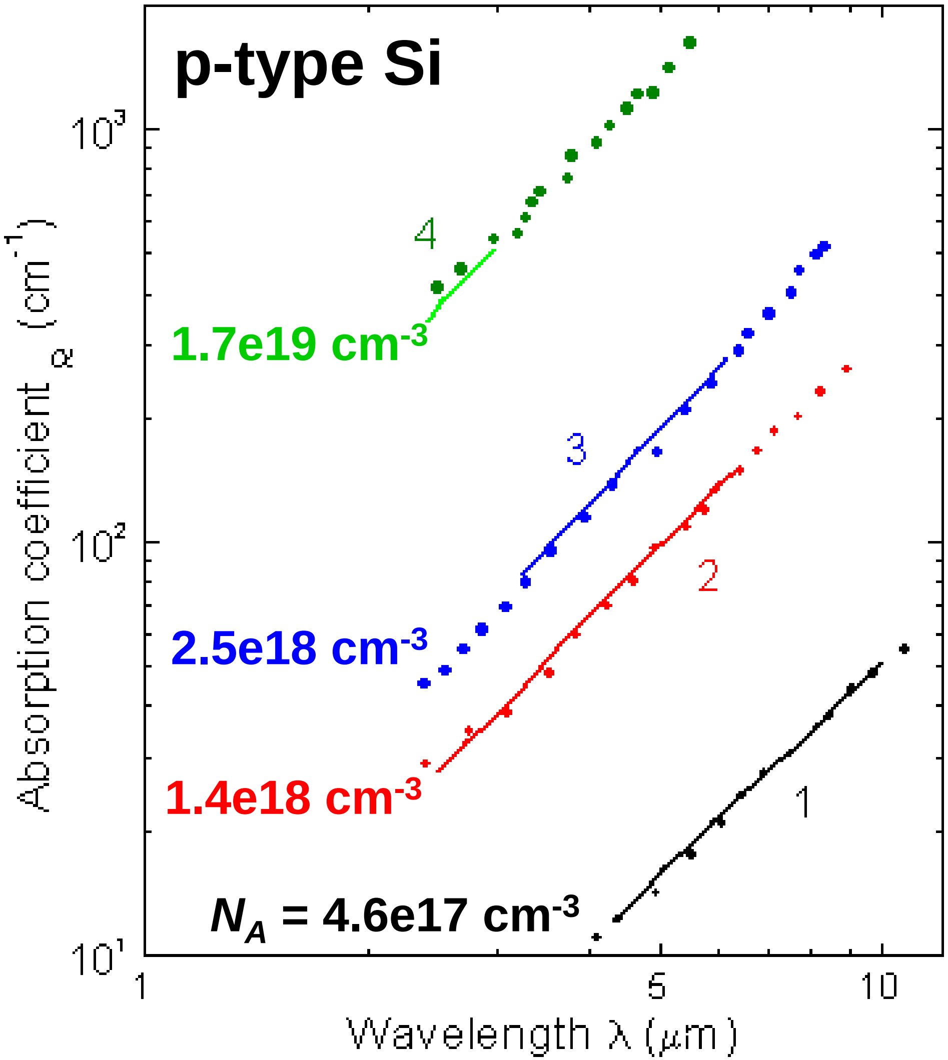
transparent in the near-infrared ($n \approx 3.5$)

absorption of optical phonons and free carriers



https://www.researchgate.net/publication/299465370_Silicon-germanium_for_photonic_applications

Example: Silicon (with dopants)

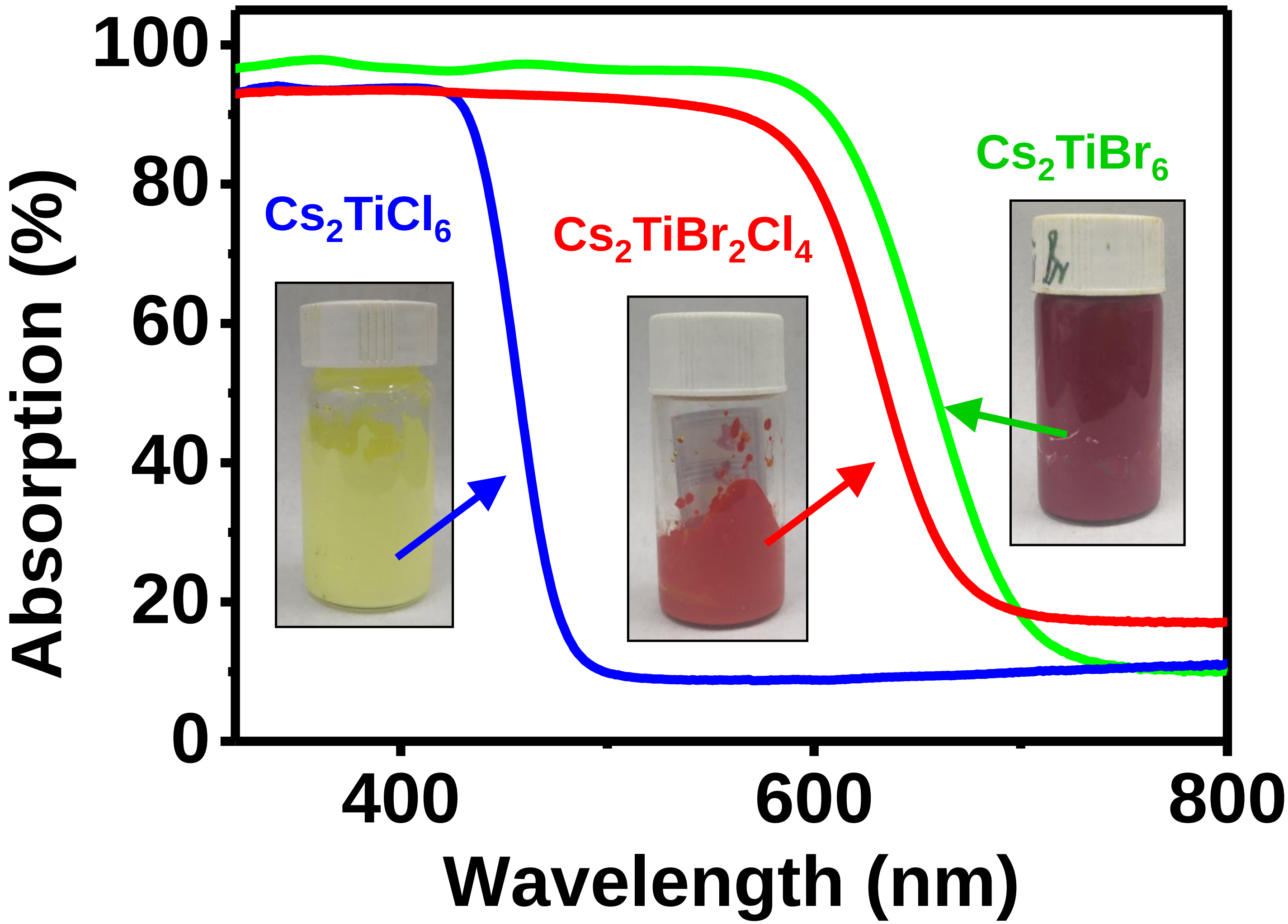


increased absorption caused by free carriers

(it can be estimated by the Drude-Lorentz Model)

H. Hara, Y. Nishi, *J. Phys. Soc. Jpn.* **21,** 1222 (1966)

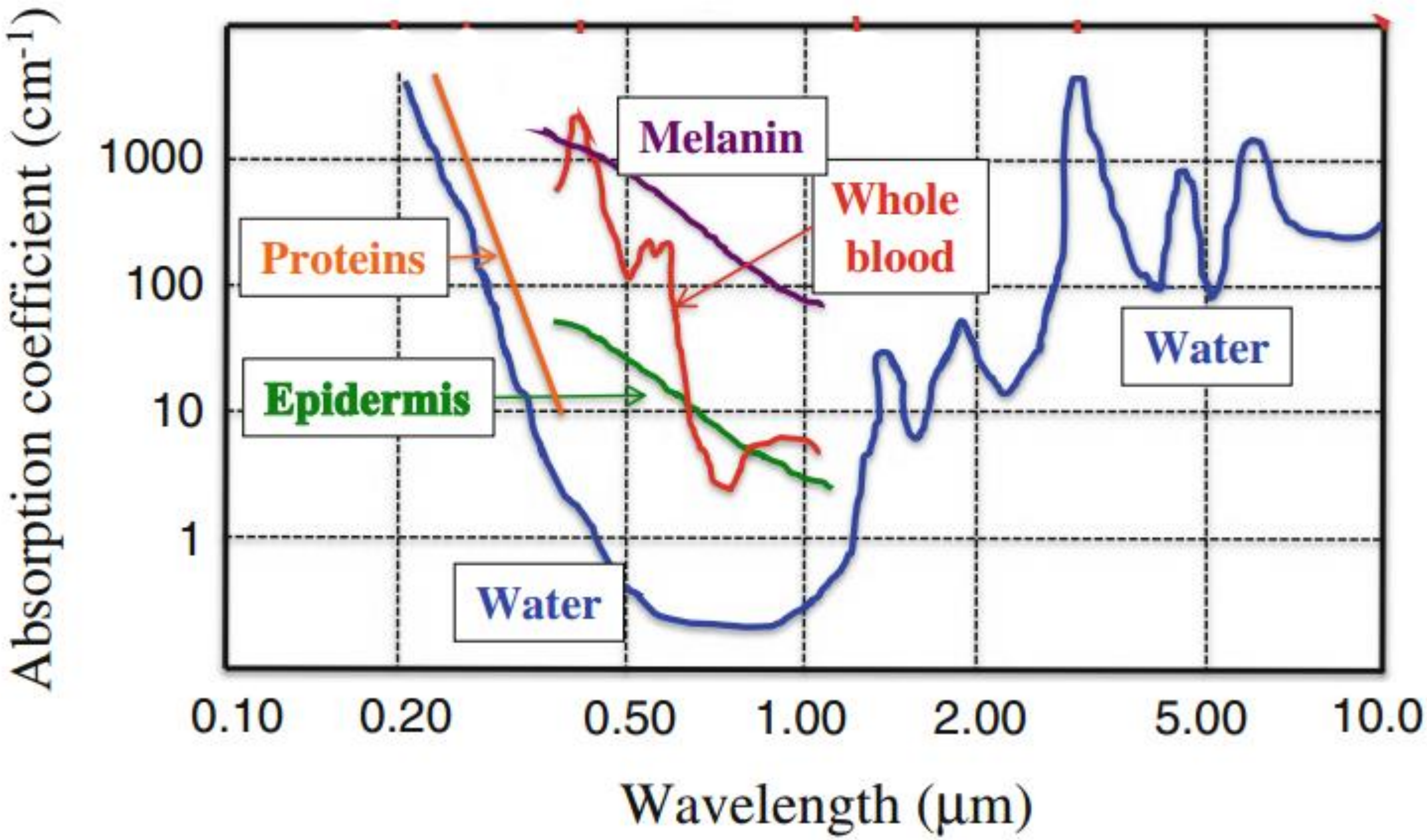
Absorption and Colors



D. Kong, *et al*, *J. Mater. Chem. C* **8,** 1591 (2020)

Absorption and Colors

Biological tissues: water, hair, skin, blood, ...

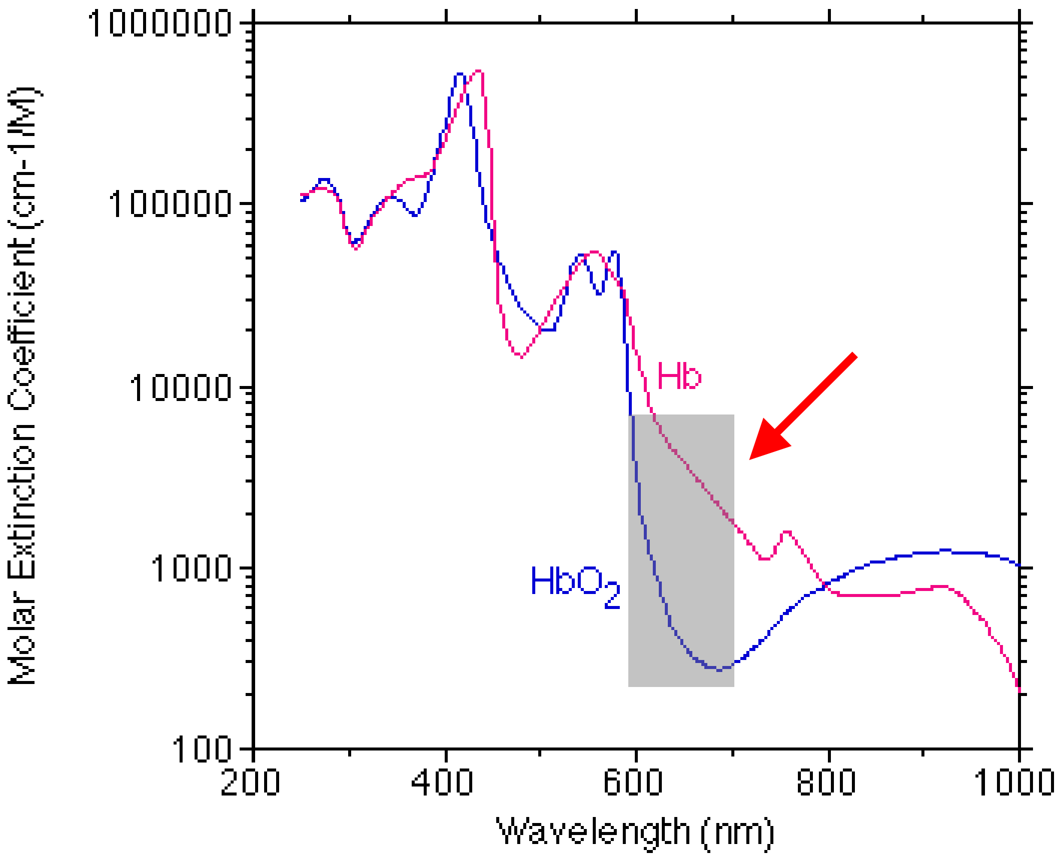


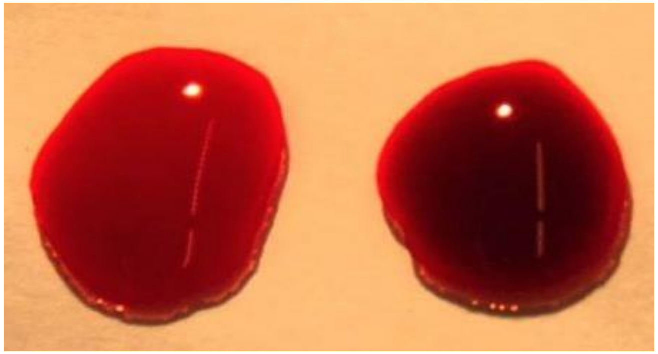
G. Keiser, *Biophotonics: Concepts to Applications* Springer**,** 158 (2016)

Example: Hemoglobin (Hb, 血红蛋白)

Hb 脱氧血红蛋白 **HbO_2 氧合血红蛋白**

Hb has higher red absorption than HbO_2





Arterial blood 动脉血 **Venous blood 静脉血**

oximeter

https://omlc.org/spectra/hemoglobin/index.html

Thank you for your attention